

ARTISTVERSE: An AI-Driven Platform for Enhancing Music Creation with ChatGPT

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Abstract— This study focuses on the development and evaluation of the "ARTISTVERSE" platform, a comprehensive AI-driven web application designed to enhance the songwriting process through four core features: Chord Progression Comparison, Song Recommendation by Mood, Dynamic Lyric Composition, and AI Lyrics Analysis. This study shows the genres' lessons learned and hearing practice from the first two features. Also discusses the unique aspects of the other two features, demonstrating ARTISTVERSE's design to transform songwriting by aligning with genres, mood, and artist styles. The application in the market review, the ARTISTVERSE designs and implementation, together with technical cooperation with ChatGPT, are proposed. Expertise perspectives for the improvement are also stated.

Keywords—Music Creation, Music Creativity, Software Engineering, Software Requirement, AI, ChatGPT

I. INTRODUCTION

In today's era, music is not only a part of daily life. It is also an important tool for building relationships, relaxing, and stimulating our creativity. Songwriting is an important process in the music industry. It is not only about creating music for entertainment purposes, but also being part of a business, media, and emotional therapy. Technology changes the way of access music. In [1], L. Liang and J. Liu analyzed how computer music production software impacts modern music composition. A comparative approach was used to assess traditional and digital composition methods. The study included technical evaluations of software features such as MIDI editing, sound synthesis, and audio-visual integration, and demonstrated that computer music production software enhances creative flexibility, efficiency, and accessibility. In [2] H. Li and F. Ren, analyze the impact of digital platforms on disseminating traditional music through qualitative data analysis and thematic exploration. The research found that "Internet + Music" has expanded the global reach of traditional music. The rapid development of artificial intelligence (AI) is also transforming the music industry, particularly in aiding musical study and the creative processes like songwriting. In [3], R. Qionggang applies digital music technology to improve music education by integrating text, images, and audio in teaching. Software such as Adobe Audition and Cubase was used for practical classroom experimentation. A comparative analysis of traditional and

technology-enhanced teaching methods was performed. Using multimedia tools significantly increased student engagement, comprehension, and creative participation. Integrating visuals and interactive elements improved music appreciation classes by making abstract concepts more relatable. In [4], B. Xu developed an AI-based intelligent music education system with modules for personalized recommendation, task management, and community interaction to recommend personalized learning content based on user preferences and progress. The system significantly increased student engagement and performance. In [5], Y. Ji and T. Chen evaluated the impact of IT-assisted teaching. The study included 12 music teachers and 239 students from middle schools. The study found that IT-assisted teaching improved music theory comprehension, sight-reading, and ear training. However, some teachers noted that excessive focus on software could distract students from music content. In [6], Y. Xiang evaluated the impact of multimedia music software in university courses. Teachers and students provided feedback on usability and learning outcomes. The findings indicated that the platform improved teaching efficiency and student engagement. In [7], J. Liu and L. Liang analyzed the implementation of computer music technology in music education by incorporating multimedia tools like sequencing software, digital instruments, and interactive learning modules to improve students' understanding of music theory and composition. Interactive software enabled more flexible learning environments, promoting creativity and collaboration. In [8], X. Ye analyzed the application of digital synthesis techniques such as spectrum, modeling, and modulation synthesis in music composition. Practical experiments were conducted using Max/MSP and other software to evaluate sound creation methods. The study highlighted the effectiveness of real-time sound synthesis in creative workflows. In [9], L. Mou et al. introduce MemoMusic 3.0, a music generation model integrated with music theory. It used Contextual information such as time, weather, and user mood to personalize music recommendations. MemoMusic 3.0 significantly enhances music recommendation accuracy and user satisfaction by considering contextual emotional states. In [10], N. Venkatachalam et. al introduce MusiciAI to interpret textual prompts and generate personalized therapeutic music. Also, in [11], X. Yan, C. M, and Y. Hao propose text-to-music generation. The approach integrates a multi-stage process that transforms symbolic MIDI sequences into audio WAV formats. Music also impacts emotional perception in both engagement and therapy. In [12], S. Yang et al. study music

emotion perception in live concerts and controlled lab environments. Participants annotated emotions in real time using a mobile app during performances to uncover cognitive factors influencing their ratings. The study found varying levels of agreement among participants, influenced by factors such as musical structure, performance expression, and individual backgrounds. In [13], J. Gu et al. conducted a comprehensive survey of studies related to the therapeutic effects of music, focusing on sleep improvement, mood regulation, and disease management. It categorized musical features and their psychological impacts through meta-analysis and clinical trial data. The survey confirmed that music interventions can improve mental health outcomes, alleviate symptoms of depression, and enhance memory and cognitive function.

ChatGPT is transforming educational practices, offering digital assistance for homework and research [14]. In [15], B. J. Carone and P. Ripollés develop SoundSignature, a music analysis web application that uses Music Information Retrieval (MIR) and Natural Language Processing (NLP) to provide users with insights about their musical preferences using OpenAI. The study demonstrated the platform's potential in providing accessible and educational music to musically knowledgeable and non-technical users. In [16], M. M. Amin et al. evaluate the performance of ChatGPT (GPT-3.5 and GPT-4) on 13 affective computing tasks, such as sentiment analysis, emotion detection, sarcasm detection, and personality assessment. and found that ChatGPT demonstrated strong performance in tasks with explicit signals but not in implicit signals. In [17], G. Rani, J. Singh, and A. Khanna proposed a comparative framework to evaluate various Generative AI models, including GANs, GPT, VAEs, and Flow-based models. They state that GPT models dominate in natural language generation.

Many musicians and songwriters often have a difficult time composing new lyrics and finding inspiration for their lyrics. Especially beginning songwriters may lack the skills necessary to compose effectively and may have difficulty creating chords for their songs due to a lack of expertise, and may also have difficulty finding new music that matches their current mood or preferences. In this work, we design a web application platform to help study and compose music by using the help of ChatGPT by proposing 4 software features: Chord Progression Comparison, Song Recommendation by Mood, Dynamic Lyric Composition, and AI Lyrics Analysis. The software helping with the music generation is reviewed and compared in part II. The details of ARTISTVERSE's inspiration and design are stated in part III. The expert discussion, conclusions, and future work are stated in part IV

II. RELATED WORK

Many websites offer online songwriting services. This work reviews 5 related websites, which are HIX.AI, Kukkik AI, Rytr, WORDHERO, and These Lyrics Do Not Exist. Pros and cons of each website are also stated and compared to ARTISTVERSE.

A. HIX.AI [18]

HIX.AI is a suite of advanced AI-based writing tools based on machine learning and natural language processing models. This website offers creative freedom and customization options for composing songs. Users can choose from various genres, target audience, and tone of voice for their lyrics. Users can choose from various languages for song lyrics. The resulting lyrics are unique, rarely repetitive, and feature connecting words. It serves as an accessible platform for both budding songwriters and established artists seeking inspiration. However, it limits the free use with a small trial period, and the subscription plans are priced at \$7.99 USD/month, \$11.99 USD/month, and \$39.99 USD/month.

B. Kuki.AI [19]

Kuki.AI is an award-winning conversational AI assistant. It is designed to be a fun and entertaining virtual companion that can chat with humans through text and voice. On this website, users can choose how long they want their lyrics to be. There are 50–150 words to choose from, and they can input the beginning of the lyrics. Users can compose songs in both Thai and English, can choose the desired word count and the desired length, and input the opening words to tailor a song. However, the lyric length options are allowed only 50, 100, and 150 words, and the meaning of the resulting song still needs improvement.

C. Rytr [20]

Rytr is an AI writer that lets you automatically generate high-quality content for various use cases. This website is a service that allows songwriters to write lyrics in their style. Users can enter song ideas for lyrics and choose the language they want. The tone of the lyrics can be selected. The user can select the number of variables and the creativity level of the lyrics. Users can compose lyrics by choosing the language, tone, and song idea of the lyrics as the user want. They can select the number of variants of the lyrics and adjust the creativity level of the lyrics, allowing the user to control the level of creativity desired. However, the resulting lyrics do not have any connecting words. It makes the lyrics look disorganized.

D. WORDHERO [21]

WORDHERO is the AI for blog writing. This website is a service that allows songwriters to compose lyrics according to their style. Users can input the short description of their song lyrics. Users can input song ideas for their lyrics. However, the options for composing songs are not diverse, and the lyrics that came out weren't very pretty. There is no connecting word in the lyrics, making the lyrics seem a bit disorganized.

E. These Lyric Do Not Exist [22]

This website is a service that allows songwriters to compose lyrics according to their style. Users can input the song topic, lyric genre, and lyric mood of the desired lyrics. Users can select the desired song topic, lyrics genre, and mood of the lyrics to get lyrics that match their needs and feelings. However, the options for composing songs are not very diverse, as they are limited to choosing the song topic, lyric genre, and lyric mood.

From review tools in the market, we proposed ARTISTVERSE to be an AI-driven platform designed to enhance the songwriter's experience by providing four core features: Chord Progression Comparison, Song Recommendation by Mood, Dynamic Lyric Composition, and AI Lyrics Analysis. Each feature enables users to personalize and enrich their music creation, in terms of musical study and generation, with the help of the Platform structures and ChatGPT. Because of the limitations of cost and time to implement a software, we propose 4 features to help learn music and compose a song. ARTISTVERSE ability, together with the comparison of applications in the market, is proposed as shown in Table I. Compared to existing solutions, ARTISTVERSE offers a holistic and integrated approach by combining the usages required by the songwriter, implemented in the different applications, into a single platform to let users experience a seamless, interactive, and personalized songwriting process.

TABLE I. SONG WRITING SERVICE WEBSITE COMPARISON

Features	A	H	K	R	W	X
Input the desired opening words or song topic for the lyrics.	✓	✓	✓	✓	✓	✓
Choose the length of the lyrics.	✓		✓			
Choose language	✓	✓	✓	✓		
Choose the genre of the lyrics.	✓	✓				✓
Choose the tone of voice or the target audience of the lyrics.		✓				
Choose a lyrical mood.	✓					✓
Choose a writing style for the lyrics according to the artist.	✓					
Recommend a song by emotion.	✓					
Chord progression	✓					
AI Lyrics Analysis	✓					

Note that A = Artistverse, H = HIX.AI, K = Kukkik AI, R = Rytr, W = WORDHERO, X = These Lyrics Do Not Exist

III. PLATFORM DESIGN AND IMPLEMENTATION

Artistverse is designed to use web application platforms to increase change for the composer to engage in musical composition and learning. Artiverse has an objective to help songwriters and musicians create unique and interesting songs. It lets users create lyrics based on their ideas by selecting the desired language in the lyrics, writing lyrics in the style of a favorite artist, choosing different moods in the lyrics, and selecting the desired music genre. This can bring personalization and relevance to a song. Artiverse uses a user-centric design by offering a less interactive interface with some controls to ensure ease of use for beginners and

experienced songwriters. The control will help interact with ChatGPT to utilize the platform efficiently and adhere to music theory principles. Real-time responsiveness of the lyric based on user inputs, the song feedback from AI, the recommended song from Spotify, and the sound from the chord progression, is given immediate feedback to the users. This can enhance users' engagement throughout the creative process. These standards collectively make ARTISTVERSE an innovative and reliable tool for AI-powered music creation. The system architecture of the platform is shown in Fig. 1.

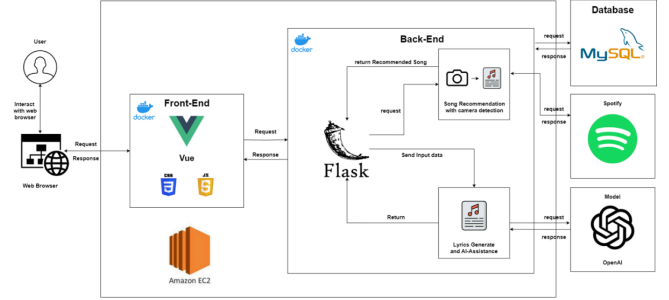


Fig. 1. System Architecture

As shown in Fig. 1, the system architecture consists of a front-end built using Vue.js, a lightweight Python web framework called Flask, and Docker for containerization. The front-end allows us to create interactive web applications using reactive data binding and components. The back end enables us to create web applications by defining routes and handling requests and responses. Docker allows us to package our application and its dependencies into a single, portable container, ensuring easy deployment and running on any platform that supports Docker. We used the OpenAI model for the function-generated lyrics and AI-Assistance to get the return result of the lyrics generated and the question-answer based on the user input. Spotify Api is for getting a list of songs to use in the song recommendation function. The MySQL database stores input, lyrics, and other relevant information, which is used to retrieve song lyrics and other data when generating recommendations. ARTISTVERSE project implement in Acer Predator Helios 300 PH317-7338 Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz 2.59 GHz using Windows 10 Home Single Language (64 Bit) RAM 32.00GB and use MacBook Air M2 Apple M2 chip MacOS Sonoma Version 14.3.1 RAM 8.00GB. ChatGPT 3.5 is used in this project. There are 4 features proposed. The designs are separated into two main objectives: Chord Progression Comparison and Song Recommendation. Mood features are used to improve the learning and listening experience. Dynamic Lyric Composition and Lyrics Analysis features are used for musical composition. The feature inspiration and objective, the details of each feature, AI, and other techniques used are stated as follows;

A. Feature #1: Chord Progression Comparison

From music theory, Chord Progression Exploration offers visual chord diagrams and audio playback, providing a practical resource for musicians exploring progression patterns across genres. It is the basis every songwriter needs to know to compose a good song. To make our target user acquainted with chord progression, ARTISTVERSE provides chord progression diagrams and visual chord progression in each song key, shows sound examples, and

shows how to play each chord visually in a chord progression by guitar or piano. This playback option allows users to both learn and creatively apply chord theory.

Details: This feature enables users to select the key and chord progression for their songs, utilizing fundamental chords in music theory. Users can input the desired key that has C, D, E, F, G, A, B, and select a chord progression sequence, such as the common progression that has 2-5-1-6, 1-6-2-5, 1-5-6-4, 1-6-4-5, 1-4-5-1, 1-4-6-5, 1-6-2-6, 6-4-3-2. This is an example. If the user inputs key C and selects progression 1-4-5-1, the system will display the chord C-F-G-C. ("1-4-5-1" in the key of C means they want to use the first (C), fourth (F), fifth (G), and first (C) chords of the C major scale in that order) based on the user's. The system then displays the corresponding chords based on the selected key and progression. Additionally, to aid users in understanding and implementing the selected chords, the system provides visual representations of chord pictures for both piano and guitar. These pictures depict the finger positions for each chord on both instruments. Furthermore, users can hear the sound of each chord. This feature allows users to audibly confirm the chord's sound, ensuring accuracy and aiding in practice and composition. The user interface (UI) design of chord progression comparison is shown in Fig. 2

AI Technique Used: Because this feature is used to provide knowledge by showing static information of chord progression theory, we do not need AI in this feature.

Other Technique: We use a normal database query to mix and match sounds to yield a chord progression in each key

Key	Progression	1	2	3	4	5	6	7
C	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	C Dm Em F G Am Bdim
D	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	D Em F#m G#m A Bm C#dim
E	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	E Gbm Abm Bbm C# Dm D#dim
F	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	F Gm Am Bb C Dm Edim
G	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	G Am Bm C D Em F#dim
A	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	A Bm Bbm C# D#m E#dim
B	2-5-1-6	1-6-2-5	1-5-6-4	1-4-5-1	1-4-6-5	1-6-2-6	6-4-3-2	B Dbm Ebm F#m G#m A#dim

Selected Chord Progression: 2-5-1-6 in Key C

Fig. 2. UI of the chord progression comparison feature

B. Feature #2: Song Recommendation by Mood

Since songs can impact emotions [12,13], learning songs in a different mood will help compose songs to serve various emotional users. Thus, the songwriter needs to improve the listening experience. Song Recommendation by Mood will recommend songs from Spotify's emotion-driven playlists. However, to let the songwriter relax after the hard work of composing songs, ARTISTVERSE uses facial recognition to detect the songwriter's mood and suggest music that matches the detected mood. Camera detects the user's emotional state through facial recognition, suggesting song options that reflect their current mood, adding an emotionally resonant layer to music creation.

Details: Our website incorporates a mood detection camera that analyzes users' facial expressions to determine their

current mood, including sad, angry, fearful, surprised, and happy. The feature includes photo capture functionality to capture the user's facial expression based on the detected mood, and the system recommends curated songs in playlists from Spotify using the Spotify API. These songs in playlists are categorized into mood-specific selections, including sad, angry, surprised, fearful, and happy playlists. The recommended playlists are updated monthly to ensure fresh content. Users can interact with the recommended playlists by clicking on a song, redirecting them to the corresponding Spotify application to listen to the selected song. Additionally, the system dynamically detects the top 5 songs from the user's selected song and displays them on our website for further exploration and enjoyment. The UI design of chord progression comparison is shown in Fig. 3

AI Technique Used: We use a technique to learn facial expressions from an image, FER-2013 from Kaggle [23], to detect user emotion

Other Technique: We use the Spotify API to get Spotify's emotion playlists according to the playlist ID prepared for each emotion.

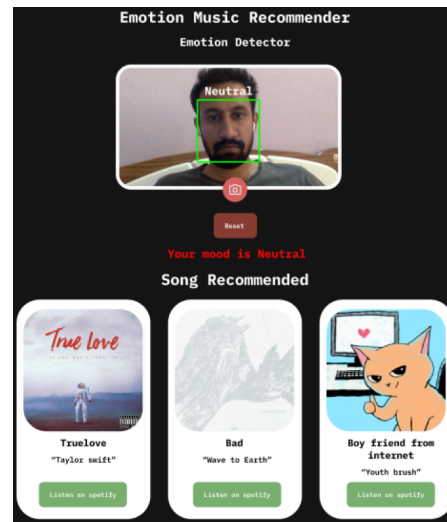


Fig. 3. UI of Song Recommendation by Mood feature

C. Feature #3: Dynamic Lyric Composition

Many tools for composing songs demonstrate how natural language processing can generate lyrics based on themes or input phrases from users. To increase accuracy, we have to state the context used [17]. The Dynamic Lyric Composition feature allows users to generate lyrics by specifying mood, genre, language, and artist style, together with the content that the user inputs. Because many songwriters may not be acquainted with working with AI, to use ChatGPT, they may forget to integrate multi-criteria inputs such as mood, genre, language, artist style, and context into ChatGPT, which can yield incorrect results. ARTISTVERSE is designed to leverage deep learning complex models by preparing text relevant to the user's specifications. However, ARTISTVERSE still keeps dynamic lyric composition styles, offering a more personalized lyric generation experience by allowing users to specify different moods, genres, languages, and stylistic preferences.

Details: This feature empowers users to generate personalized song lyrics tailored to various criteria,

including Mood, Genres, Languages, and Artist Styles. Users can input prompt words into our system and select mood, genres, languages, or even the style of their favorite artists, and the system will compose lyrics accordingly. Users can select from various mood tones, including happy, sad, romantic, inspirational, energetic, chill, hopeful, reflective, mysterious, dramatic, nostalgic, playful, confident, dreamy, and empowering. Additionally, users can specify the genre they desire, choosing from options that include pop, rock, country, R&B, hip-hop, folk, indie, electronic jazz, blues, reggae, classical, alternative, gospel, and punk. Moreover, users can select the language they want the lyrics to be composed, with options ranging from Thai, Japanese, Chinese, and Korean. Furthermore, users can compose lyrics in the style of their favorite artists, including international stars like Taylor Swift, Billie Eilish, Ed Sheeran, Justin Bieber, The Weeknd, Maroon 5, Olivia Rodrigo, The 1975, Joji, Keshi, and Thai artists like Potato, Palmy, Big Ass, Lula, Lipta, Scrubb, Stamp, Getsunova, and Da Endorphine. By providing input prompts, users can initiate the lyric generation process, allowing the system to create unique and personalized lyrics that align with their preferences, emotions, language choices, genre preferences, or favorite artist styles. Users can save the generated lyrics to their computer in a text file. The user interface (UI) design of chord progression comparison is shown in Fig. 4

AI Technique Used: We prepared text relevant to the user's input information before sending it to ChatGPT to increase accuracy [17]. The example of the prepared text is "generate lyrics that have a song topic {prompt} in {language} language and the genre is {genre}. (just give me only lyrics)"

Other Technique: We designed the UI to control information that users have to input, so the information sent to ChatGPT is enough to return a correct result. Also, because the result from ChatGPT always changes, we add a save function to let the songwriter keep the song result they like.

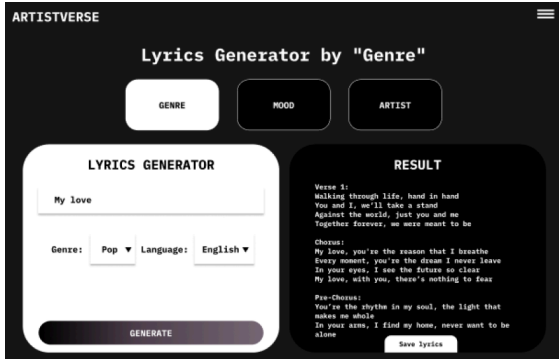


Fig. 4. UI of the dynamic lyric composition feature

C. Feature #4: AI Lyrics Analysis

Because there is no sound in the return result of ChatGPT, the AI Lyrics Analysis feature is designed to assist users in aligning lyrical content with suitable musical elements, allowing more cohesive and harmonized compositions. ARTISTVERSE analyzes lyrical content to suggest relevant musical features, like keys and chord progressions, to enhance the coherence between lyrics and music.

Details: The AI Lyrics Analysis feature analyzes user-input lyrics to suggest chords, tempo, time signature, keys, and

extracts musical information based on the provided lyrics, guiding users creatively. Users need to input lyrics only, and the system will provide a detailed musical analysis, including chord suggestions, tempo recommendations, time signature, and key identification based on their input lyrics. The user interface (UI) design of chord progression comparison is shown in Fig. 5

AI Technique Used: We prepared text relevant to the user's input information before sending it to ChatGPT to increase accuracy [17]. The example of the prepared text is "Based on my lyrics {lyrics}, can you provide a suggestion of chord, keys, time signature, tempo, and extract musical information for me that is related to my lyrics?"

Other Technique: Because users need to compare the song to the suggestions, we design an interface to show both information on the same page. Also, because the result from ChatGPT always changes, we add a save function to let the songwriter keep the song result they like.

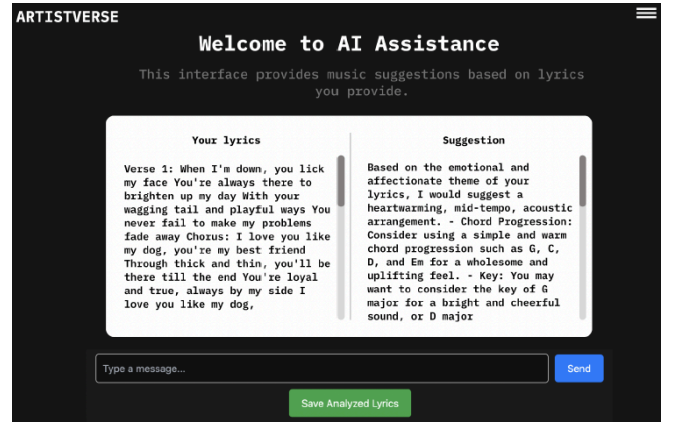


Fig. 5. UI of the AI lyrics analysis feature

IV. DISCUSSION AND CONCLUSIONS

This study proposes the development and evaluation of the ARTISTVERSE platform, an AI-powered tool that enhances songwriting through four integrated features: Chord Progression Comparison, Song Recommendation by Mood, Dynamic Lyric Composition, and AI Lyrics Analysis. UI and AI techniques are also stated. However, after an interview with 3 experts who have experience playing music and composing songs, even though they were satisfied with our proposed application, they suggested adding more features as follows;

- Because the results of the recommendation of the song are limited to the completeness of the question stated to ChatGPT and the ChatGPT database itself, the prepared text before sending to ChatGPT needs to be improved to gain more accuracy.
- Real-time edit feature is required. The composer needs to edit songs with the help of a system.
- Composers need to hear the sound of the song while they are composing.
- Composers need to export the song in five-line staff, so they can share the notes with the band easily.

Other than the suggestion to the expert, we plan to do the usability test of Artiverse from the perspective of musical

learning and composing a song. We will interview more target users together with seeing their behavior while using ARTISTVERSE to find more requirements to complete the cycle of composing a song to support every level of songwriters.

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